

MATH 345 Skeleton Notes

Unit 5: Optimization and training

Section 5.4: Symmetric matrices and quadratic forms

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In this section, we discuss diagonalization for symmetric matrices, which were defined in Unit 1. Symmetric matrices play a fundamental role in optimization because Hessian matrices are symmetric when the mixed second partial derivatives agree. Another recurring example is $A^T A$, which appeared in the Unit 4 normal equations and will return below in least squares.

Orthogonality of eigenvectors

A key property of symmetric matrices is that for any two vectors $\mathbf{x}$ and $\mathbf{y}$, $(A\mathbf{x}) \cdot \mathbf{y} = \mathbf{x} \cdot (A\mathbf{y})$. Indeed, we calculate that

$$\begin{aligned}(A\mathbf{x}) \cdot \mathbf{y} &= (A\mathbf{x})^T \mathbf{y} \\ &= \mathbf{x}^T A^T \mathbf{y} \\ &= \mathbf{x}^T A \mathbf{y} \quad (\text{since } A^T = A) \\ &= \mathbf{x} \cdot (A\mathbf{y})\end{aligned}$$

This property leads to a fundamental result about the eigenvectors of symmetric matrices.

Theorem: Orthogonality of Eigenvectors If A is a *symmetric matrix*, then the *eigenvectors* of A corresponding to *distinct* eigenvalues are *orthogonal*.

Proof Let $A\mathbf{x} = \lambda\mathbf{x}$ and $A\mathbf{y} = \mu\mathbf{y}$, where $\lambda \neq \mu$. Using the property above:

$$\begin{aligned}\lambda(\mathbf{x} \cdot \mathbf{y}) &= (\lambda\mathbf{x}) \cdot \mathbf{y} \\ &= (A\mathbf{x}) \cdot \mathbf{y} \\ &= \mathbf{x} \cdot (A\mathbf{y}) \\ &= \mathbf{x} \cdot (\mu\mathbf{y}) \\ &= \mu(\mathbf{x} \cdot \mathbf{y})\end{aligned}$$

This gives us $(\lambda - \mu)(\mathbf{x} \cdot \mathbf{y}) = 0$. Since $\lambda \neq \mu$, we must have $\mathbf{x} \cdot \mathbf{y} = 0$, which means the eigenvectors are orthogonal.

As we will see, it turns out that for any symmetric matrix, there exists a pairwise orthogonal set of eigenvectors which form a basis of $\mathbb{R}^n$. In particular, every symmetric matrix is diagonalizable.

Orthogonal matrices and orthogonal diagonalization

Recall the definition of an *orthogonal matrix* from Orthogonal Matrices.

Theorem: Equivalent Conditions for Orthogonal Matrices Let Q be an $n \times n$ matrix. Then the following conditions are equivalent to one another:

1. Q is invertible and $Q^{-1} = Q^T$.
2. The rows of Q are orthonormal.
3. The columns of Q are orthonormal.

Proof Let us start by showing the first property is equivalent to the third. Suppose Q is invertible, and $Q^{-1} = Q^T$. Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be the columns of Q . Recall from Theorem thm-3-3-matrix-multiplication-dot-product that the (i, j) -entry of $Q^T Q$ is equal to the dot product $\mathbf{v}_i \cdot \mathbf{v}_j$. Since $Q^{-1} = Q^T$, $Q^T Q = I$, and so for $i \neq j$, $\mathbf{v}_i \cdot \mathbf{v}_j = 0$, and for each i , $\mathbf{v}_i \cdot \mathbf{v}_i = 1$. Thus

the column vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ are orthonormal. Conversely, if the column vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ are orthonormal, then we immediately see that $Q^T Q = I$.

The proof that the first property is equivalent to the second is very similar, using the rows of Q rather than the columns and that if $Q^T Q = I$, then $Q Q^T = I$, and vice versa.

Lemma Let Q be an $n \times n$ orthogonal matrix. Then:

1. Q preserves dot products, i.e. for any vector $\mathbf{v} \in \mathbb{R}^n$, $(Q\mathbf{v}) \cdot (Q\mathbf{w}) = \mathbf{v} \cdot \mathbf{w}$.
2. Q preserves lengths, i.e. for any vector $\mathbf{v} \in \mathbb{R}^n$, $\|Q\mathbf{v}\| = \|\mathbf{v}\|$.

Proof Let $\mathbf{v}$ and $\mathbf{w}$ be column vectors. Recalling the fifth property of Properties of matrix multiplication, we calculate that

$$\begin{aligned} (Q\mathbf{v}) \cdot (Q\mathbf{w}) &= (Q\mathbf{v})^T (Q\mathbf{w}) \\ &= \mathbf{v}^T Q^T Q \mathbf{w} \\ &= \mathbf{v}^T I_n \mathbf{w} \\ &= \mathbf{v}^T \mathbf{w} \\ &= \mathbf{v} \cdot \mathbf{w}. \end{aligned}$$

Thus Q preserves dot products.

Next, using that Q preserves dot products, and also Remark rem-1-1-length-as-dot-product, we find

$$\|Q\mathbf{v}\|^2 = (Q\mathbf{v}) \cdot (Q\mathbf{v}) = \mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2.$$

Taking square roots on both sides of this equation gives $\|Q\mathbf{v}\| = \|\mathbf{v}\|$.

textbook figure fig-orthogonal-transformation provides a visualization of an orthogonal change of coordinates.

Definition: Orthogonal diagonalizability A matrix A is

orthogonally diagonalizable

when an *orthogonal* matrix Q can be found such that $Q^{-1}AQ$ is a diagonal matrix. Since $Q^{-1} = Q^T$, this is equivalent to $Q^T AQ$ being a diagonal matrix.

Activity

Consider the symmetric matrix

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

Check that the vectors

$$\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

are eigenvectors of A which are orthogonal to each other. What are the corresponding eigenvalues?

textbook figure fig-matrix-transformation should be compared to fig-7-1-eigenvector-transformation. Notice that since the matrix A in Activity ex-symmetric-matrix-2x2 is symmetric, the red and blue line in textbook figure fig-matrix-transformation meet at right angles, since the vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, whereas the red and blue lines in fig-7-1-eigenvector-transformation do not meet at right angles, since the eigenvectors for that example are not orthogonal to one another.

Activity

Consider the matrix

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Find an orthogonal matrix Q such that $Q^{-1}AQ$ is diagonal.

Activity

Suppose that A is orthogonally diagonalizable, i.e. we can find an orthogonal matrix Q so that $Q^T A Q = D$, where D is diagonal. Then $A = Q D Q^T$.

Remark The factorization $A = Q D Q^T$ writes A as a product of three matrices: an orthogonal matrix, a diagonal matrix, and the transpose of the orthogonal matrix. This product is sometimes called the

spectral decomposition

of a symmetric matrix A .

Principal axes theorem

Recall Theorem thm-7-1-diagonalizability. That theorem tells us that if A is orthogonally diagonalizable, where $Q^{-1}AQ$ is diagonal, then the columns of Q must be linearly independent eigenvectors of A . In addition, in order for Q to be orthogonal, Equivalent Conditions for Orthogonal Matrices tells us that the eigenvectors which form the columns of Q must be orthonormal.

Remarkably, *all* symmetric matrices are orthogonally diagonalizable.

Theorem: Principal Axes Theorem Let A be any $n \times n$ matrix. Then the following conditions are equivalent:

1. A has an orthonormal set of n eigenvectors
2. A is orthogonally diagonalizable
3. A is symmetric

A set of orthonormal eigenvectors for a symmetric matrix A is called a set of

principal axes for A .

Proof We do not prove the “hard” part of the theorem (that the third property implies the second). But the proof that the second property implies the third is more simple (see Activity ex-orthogonal-diag-implies-symmetric).

Remark Principal Axes Theorem is also called *The Spectral Theorem* for symmetric matrices (the set of eigenvalues of a matrix is often called the **spectrum** of the matrix).

We do not prove the “hard” part of this theorem, that is, that (3) implies (2). Let’s however check that (2) implies (3).

Activity

Show that if A is orthogonally diagonalizable, then A is symmetric.

Activity

Let

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ -1 & 2 & 5 \end{bmatrix}$$

Find an orthogonal matrix Q such that $Q^{-1}AQ$ is diagonal.

In Activity act-3x3-orthogonal-diagonalization, the eigenvalues of A were distinct. To come up with a general strategy to obtain an orthogonal diagonalization, we need to make one further observation.

Definition: Eigenspaces of a matrix Suppose A is a square matrix. Then for any scalar λ , the set

$$E_{\lambda}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \lambda\mathbf{x}\}$$

is called the **eigenspace** of A corresponding to λ .

Theorem Suppose A is a square matrix and $\lambda \in \mathbb{R}$. Then $E_\lambda(A)$ is a subspace of $\mathbb{R}^n$.

Proof Note that $\mathbf{x} \in E_\lambda(A)$ precisely when $\mathbf{x}$ solves the equation $(A - \lambda I)\mathbf{x} = \mathbf{0}$. Thus $E_\lambda(A)$ is precisely the null space of the matrix $A - \lambda I$, and the theorem follows from Theorem thm-5-1-null-space-is-a-space.

Note: Procedure for orthogonally diagonalizing a symmetric matrix A

1. Find the eigenvalues of A .
2. For each eigenvalue λ of A , find a basis for $E_\lambda(A)$, and convert it to an orthonormal basis using the Gram-Schmidt orthogonalization process (see The Gram-Schmidt orthogonalization algorithm) if necessary.
3. Collecting the orthonormal bases for each of the eigenspaces gives an orthonormal basis of eigenvector for A . Taking these eigenvectors as column vectors of a matrix Q gives an orthonormal matrix such that $Q^{-1}AQ$ is diagonal.

Activity

If

$$A = \begin{bmatrix} 8 & -2 & 2 \\ -2 & 5 & 4 \\ 2 & 4 & 5 \end{bmatrix},$$

compute an orthonormal diagonalization of A . You may use the fact that the characteristic polynomial of A is $\lambda(\lambda - 9)^2$, and the null space of A is spanned by the vector

$$\begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}.$$

Definition: Positive definite, negative definite, and indefinite matrices A symmetric matrix is called

positive definite

if all of its eigenvalues are strictly positive. It is called

negative definite

if all of its eigenvalues are strictly negative. It is called

indefinite

if it has at least one strictly positive eigenvalue and at least one strictly negative eigenvalue.

The fact that every symmetric matrix is diagonalizable can often be used to determine structural properties of such matrices. For instance, consider the theorem below.

Theorem A symmetric matrix C is positive definite if and only if all eigenvalues of C are strictly positive.

Proof By Principal Axes Theorem, we can write $C = QDQ^T$, where Q is an orthogonal matrix, and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ is a diagonal matrix. Then if $\mathbf{y} = Q^T \mathbf{x}$,

$$\mathbf{x}^T C \mathbf{x} = \mathbf{x}^T (Q D Q^T) \mathbf{x} = \mathbf{y}^T D \mathbf{y} = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2.$$

This quantity is positive for all $\mathbf{x} \neq 0$, or equivalently, since Q^T is invertible, for all $\mathbf{y} \neq 0$, if and only if $\lambda_1, \dots, \lambda_n > 0$.

Quadratic forms and definiteness

Definition The

quadratic form

associated with an $n \times n$ matrix A is the scalar-valued function Q_A on $\mathbb{R}^n$ defined by $Q_A(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$.

Activity: A quadratic form as curvature

Let

$$H = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}.$$

1. Compute $\mathbf{h}^T H \mathbf{h}$ for

$$\mathbf{h} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{h} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{h} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

2. Write $\mathbf{h}^T H \mathbf{h}$ for a general vector

$$\mathbf{h} = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}.$$

3. In which coordinate direction is the quadratic form larger?

Tags. [U5-L04, U5-L05 | P+C | Core]

If we can understand quadratic forms, we can understand the quadratic approximations of functions at critical points.

Activity

For a symmetric 3×3 matrix $A = [A_{ij}]$, write out the quadratic form $q_A(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ explicitly in terms of the entries of A and the components of $\mathbf{x} = (x_1, x_2, x_3)$.

Remark More generally, for a symmetric $n \times n$ matrix $A = [A_{ij}]$, the quadratic form $q_A(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ can be written as:

$$q_A(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} = \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j = \sum_{i=1}^n A_{ii} x_i^2 + 2 \sum_{1 \leq i < j \leq n} A_{ij} x_i x_j$$

In particular, note that if A is a diagonal matrix, then

$$q_A(\mathbf{x}) = A_{11} x_1^2 + \cdots + A_{nn} x_n^2$$

which is simple to understand. We will use spectral theory to show that a rotation of a coordinate system reduces a quadratic form to a quadratic form defined by a diagonal matrix. The signs of the diagonal entries will then tell us the behavior of the critical point.

Recall the spectral theorem, i.e. Principal Axes Theorem, which says that there is an orthogonal matrix Q so that $Q^T A Q$ is a diagonal matrix D .

Theorem: Diagonalization Theorem Consider a quadratic form Q_A , defined in terms of a symmetric matrix A . Suppose Q is an orthogonal matrix such that

$$Q^T A Q = D = \text{diag}(\lambda_1, \dots, \lambda_n).$$

If we set $\mathbf{x} = Q\mathbf{y}$, then

$$Q_A(\mathbf{x}) = \lambda_1 y_1^2 + \dots + \lambda_n y_n^2.$$

Proof Since $\mathbf{x} = Q\mathbf{y}$, we have

$$\begin{aligned} Q_A(\mathbf{x}) &= \mathbf{x}^T A \mathbf{x} \\ &= (Q\mathbf{y})^T A (Q\mathbf{y}) \\ &= \mathbf{y}^T Q^T A Q \mathbf{y} \\ &= \mathbf{y}^T D \mathbf{y} \\ &= \lambda_1 y_1^2 + \dots + \lambda_n y_n^2. \end{aligned}$$

Definition: The principal axes If A is a symmetric, $n \times n$ matrix, then an orthonormal basis of eigenvectors of A is called a set of

principal axes

for the quadratic form Q_A .

Activity

Consider the quadratic form $q(x, y) = 13x^2 - 6xy + 5y^2$.

Task Find the symmetric matrix A such that $q = q_A$, i.e. such that

$$q(x, y) = \begin{bmatrix} x & y \end{bmatrix} A \begin{bmatrix} x \\ y \end{bmatrix}.$$

Task Find the eigenvalues and a basis of eigenvectors for A .

Task (c) Apply the previous theorem to express $q(x_1, x_2)$ in terms of new coordinates y_1 and y_2 that eliminate the cross term.

Activity

Consider the quadratic form

$$q(x, y) = x^2 + 4xy - 2y^2.$$

Task Find the symmetric matrix A such that $q = q_A$.

Task Find the eigenvalues and corresponding eigenvectors of A .

Task Apply the previous theorem to express $q(x_1, x_2)$ in terms of new coordinates y_1 and y_2 that eliminate the cross term, and classify the level

curves $q(x_1, x_2) = c$ for different values of c .

Theorem A symmetric matrix A is positive definite if and only if $\mathbf{x}^T A \mathbf{x} > 0$ for every vector $\mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$. It is negative definite if and only if $\mathbf{x}^T A \mathbf{x} < 0$ for every vector $\mathbf{x} \neq \mathbf{0}$.

Proof This follows from Diagonalization Theorem.